

Ahsan Tahir

Python Developer | AI Engineer

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Profile

AI Engineer specializing in Computer Vision and MLOps. Proven track record of deploying YOLO models to production using FastAPI and Docker. Experienced in building complex compliance logic for the Retail Tech industry.

Professional Experience

Python/AI Engineer

05/2024 – present

Concave Tech

Retail Planogram Compliance System:

- Engineered a computer vision pipeline using **YOLO & Geometric Algorithms** to automate shelf compliance checks (Horizontal/Vertical matching).
- Improved complex logic for **Brand Blocking** detection and **Eye-Level (Golden Zone)** analysis, directly impacting retail audit efficiency.
- Architected an interactive audit dashboard using **Gradio** for visualizing gaps and misplacements.

Real-Time Object Detection API & MLOps:

- Built and deployed a production-grade **FastAPI** backend for real-time inference.
- Implemented **Docker** for containerization, handling image builds, tagging, and server-side deployment on Linux environments.
- streamlined large-scale datasets using **AWS CLI** with S3-compatible object storage (Wasabi).

AI-Powered Clustering & Route Optimization System:

A full-stack web application that intelligently clusters geographic locations and optimizes daily shop visit schedules. The system uses advanced machine learning algorithms to:

- Intelligent Geographic Clustering:** Uses constrained K-means clustering to group shops by location while respecting business constraints (maximum visits per cluster, distance limits from cluster centers, time availability).
- Daily Schedule Optimization:** Distributes shop visits across multiple days while balancing workload, maintaining visit frequency requirements, and maximizing time utilization.
- Route Optimization:** Calculates geodesic distances between locations and optimizes travel routes using algorithms like heap-based path finding.
- Enterprise Features:**
 - User authentication and session management with MySQL database.
 - Excel file import/export for bulk data handling.
 - Interactive web dashboard with real-time data visualization.
 - REST API with CORS support for external integrations.
 - Performance-optimized processing using Numba JIT compilation.

Education

University of the Punjab

2019 – 2023

Bachelor of Science in Information Technology (BS-IT)

Skills

Languages: Python, Java, JS, React, C++

Frameworks & Libraries: PyTorch, YOLO, TensorFlow Lite, OpenCV, NLP, Pandas, Geopy, PyQt, Scikit-learn, Gradio

Backend: FastAPI, Flask

Web Technologies: HTML5, CSS3

Databases: MySQL, SQLite, Firebase

Cloud & DevOps: AWS, Docker, Git, Wasabi, S3-compatible storage